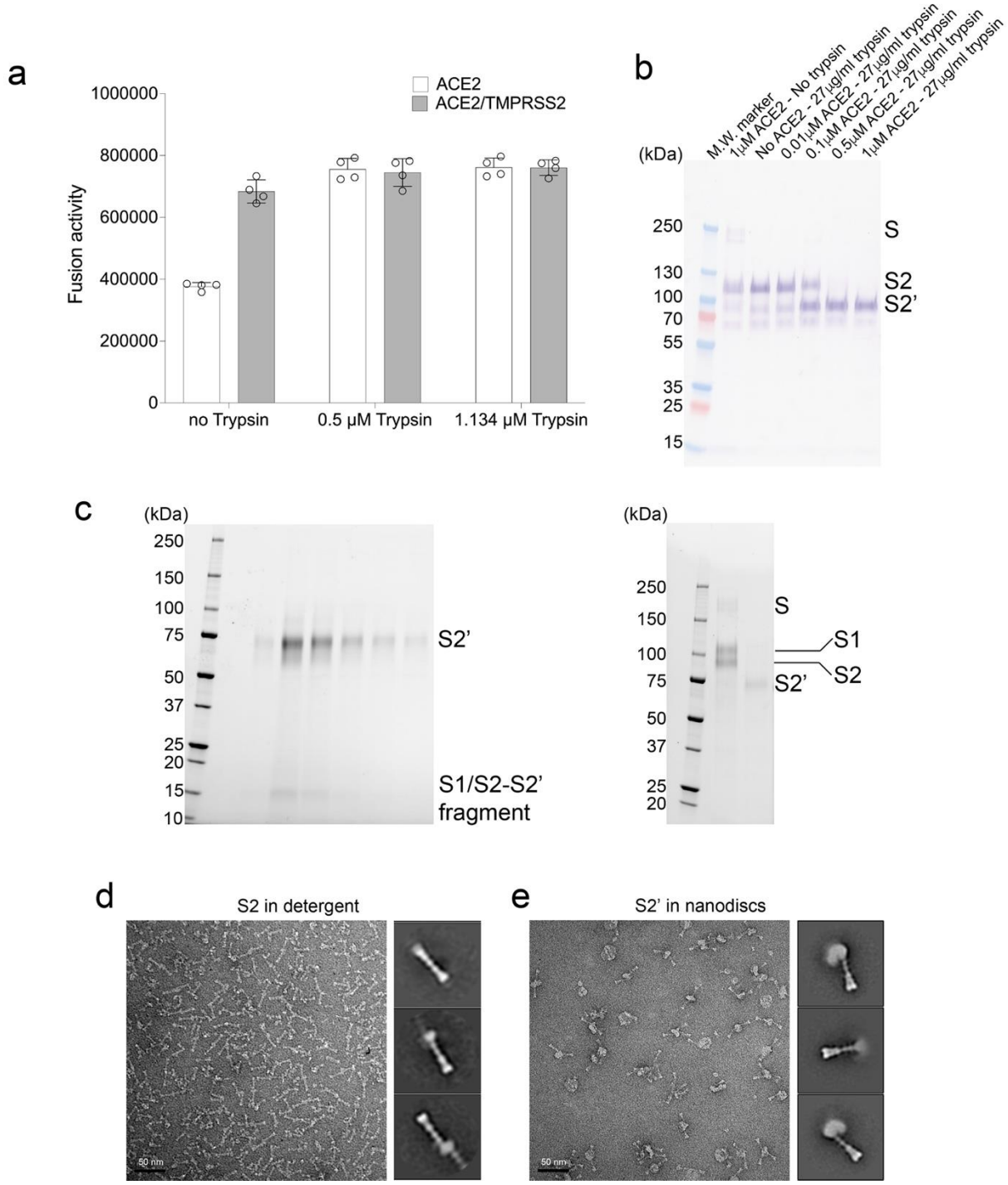
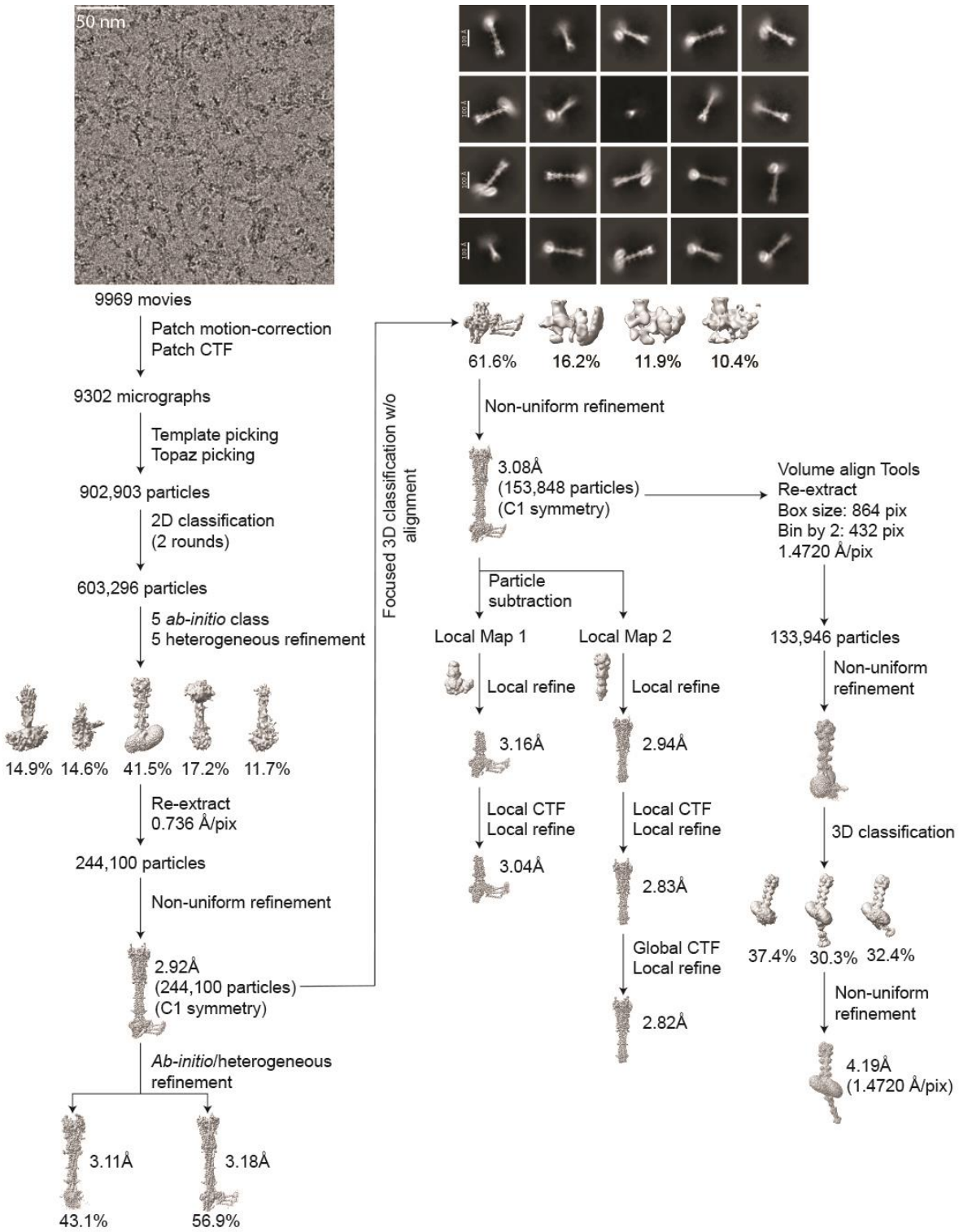
Supplementary Information

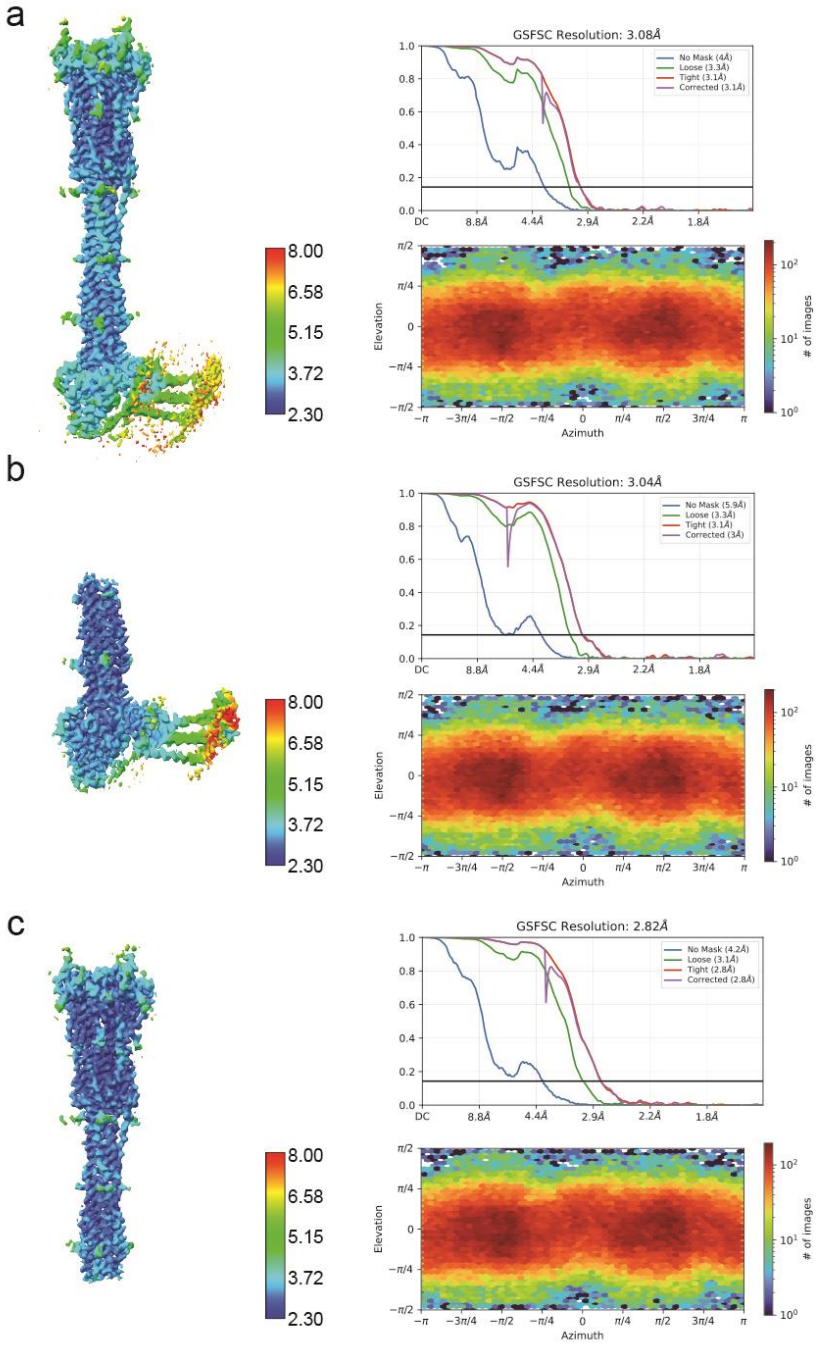


Supplementary Figure 1. Production of the S2' fragment of SARS-CoV-2 spike. (**a**) Both TMPRSS2 and trypsin enhance S-mediated membrane fusion. HEK293T cells transfected with the full-length S protein (G614) were tested for membrane fusion with the ACE2-expressing cells or ACE2- and TMPRSS2-expressing cells in the presence of trypsin. The experiment was repeated

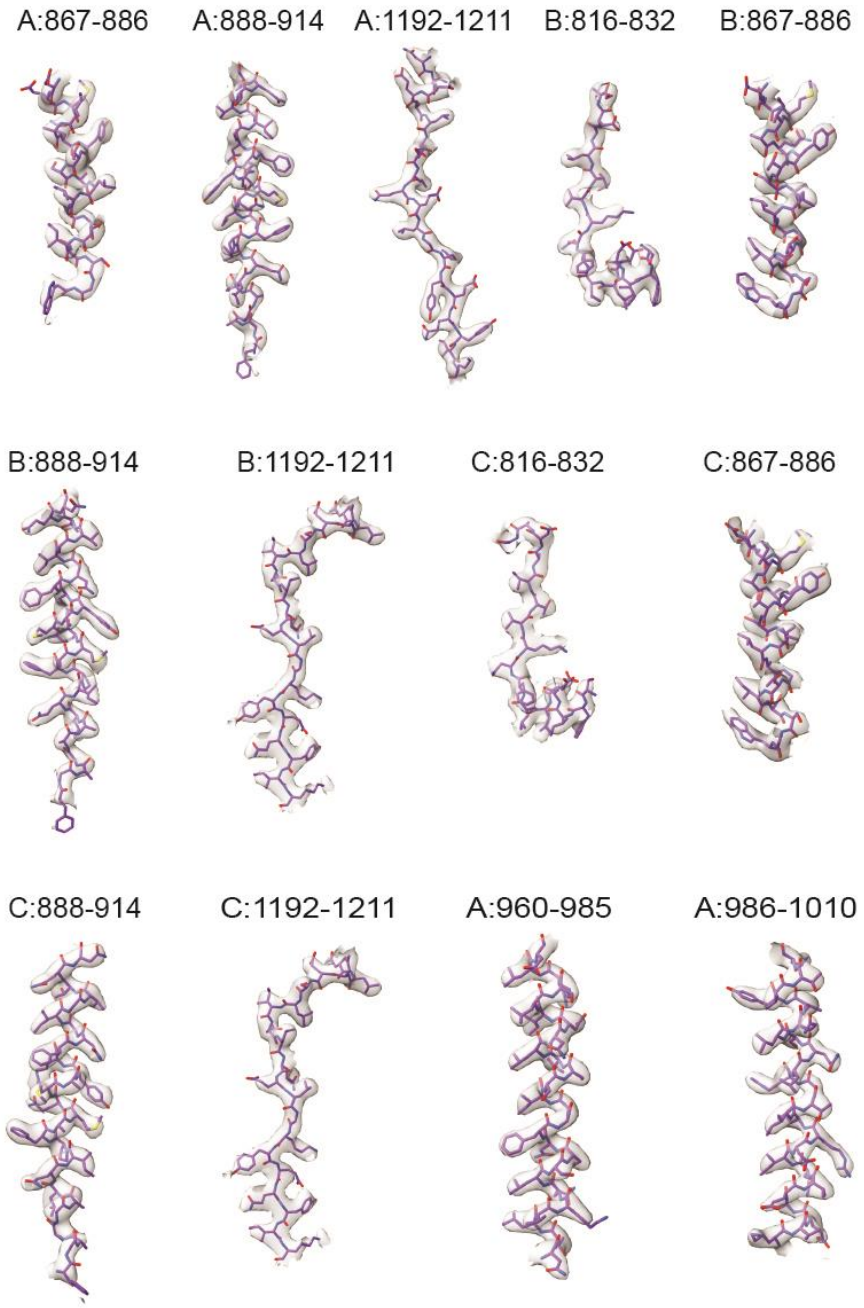
at least three times with similar results. (**b**) Cleavage of the S protein by trypsin in the presence of ACE2. S-expressing cells were treated with soluble ACE2 and trypsin at various concentrations as indicated and their lysates were analyzed by western blot using anti-strep tag antibody. (**c**) Left, SDS-PAGE analysis of the fractions eluted from the strep-tactin column for purifying the S2' fragment. The S1/S2-S2' fragment with a calculated size of 14 kDa migrated at ~15 kDa. Right, SDS-PAGE analysis showing the purified S2' fragment migrated faster than the uncleaved S2, as expected. (**d**) Negative stain EM of the uncleaved S2 trimers in detergent showing clusters of trimers. Representative image and 2D averages are shown, and the box size of 2D averages is ~460Å. (**e**) Negative stain EM of the S2' trimers reconstituted in nanodiscs showing clusters of trimers. Representative image and 2D averages are shown, and the box size of 2D averages is ~460Å.



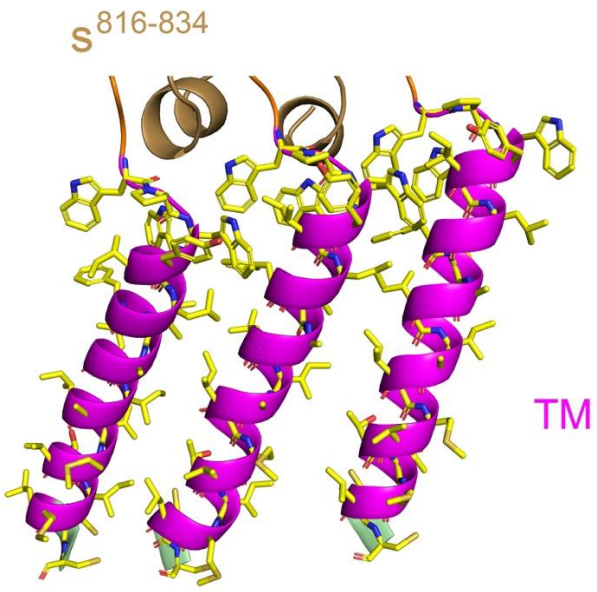
Supplementary Figure 2. Cryo-EM analysis of the S2' trimer. Top, representative micrograph out of 9,969 similar micrographs**,** and 2D averages of the cryo-EM particle images of the S2' trimer in detergent. Bottom, data processing workflow for structure determination.



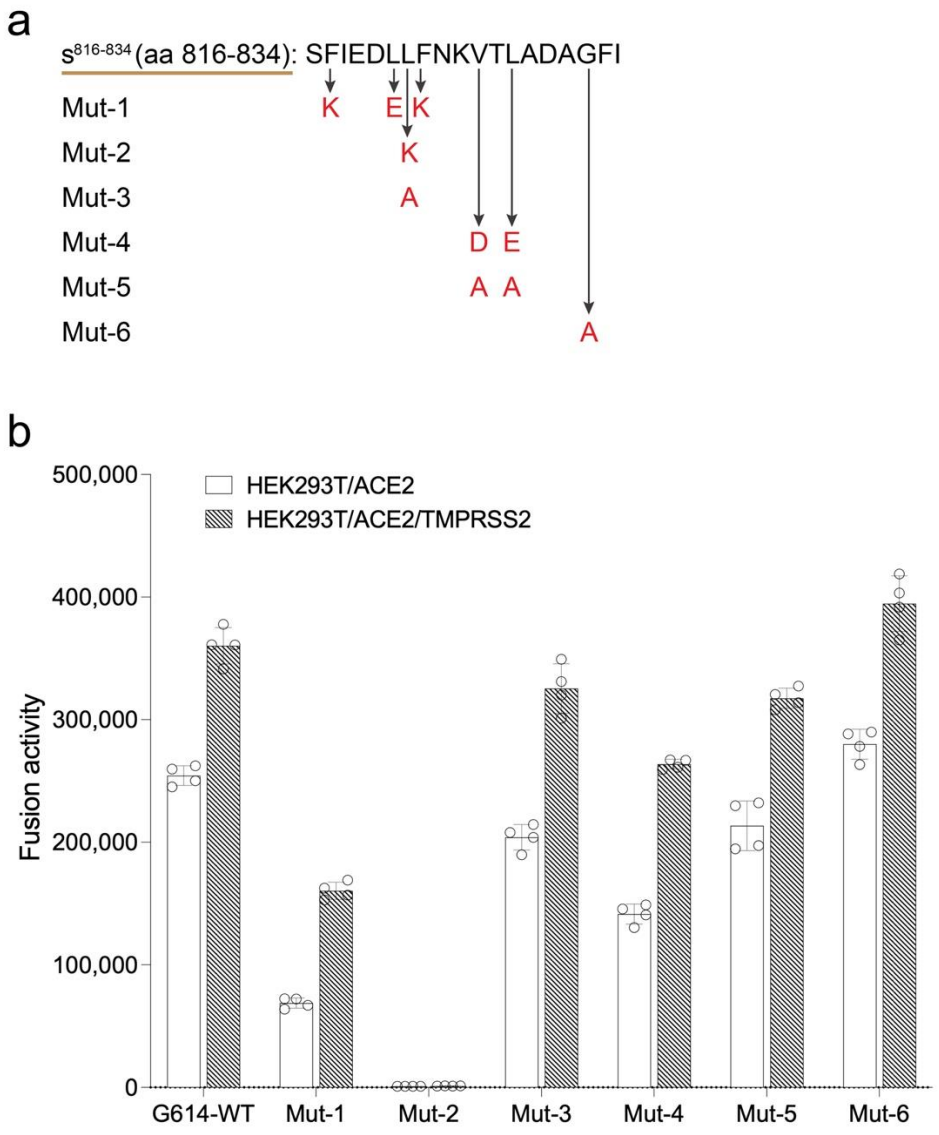
Supplementary Figure 3. Analysis of the S2' trimer structures by cryoSPARC. (**a**)-(**c**) 3D reconstructions of the S2' trimer from the overall refinement and local refinement (top and bottom) are colored according to local resolution estimated by cryoSPARC with the FSC=0.5 criterion. Gold standard FSC curves of the three refined 3D reconstructions of the S2' trimer and the corresponding cryoSPARC output for particle distribution of each reconstruction.



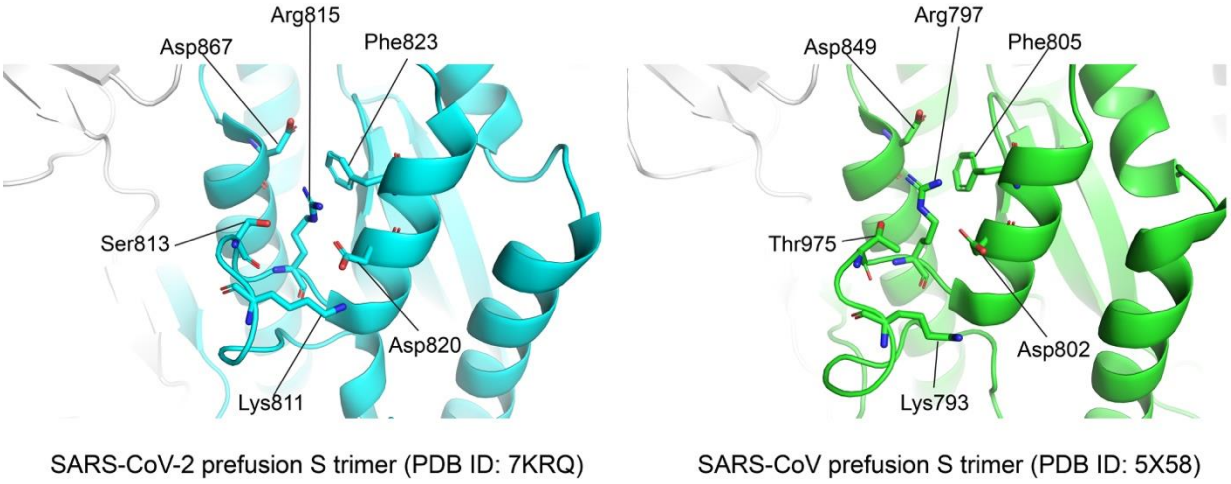
Supplementary Figure 4. Additional analysis of the S2' trimer structure. Representative density in gray surface representation from the EM map of the S2' trimer. The model range is indicated by chain name and residue numbers. For example, A:867-886 represents chain A: residues 867-886.



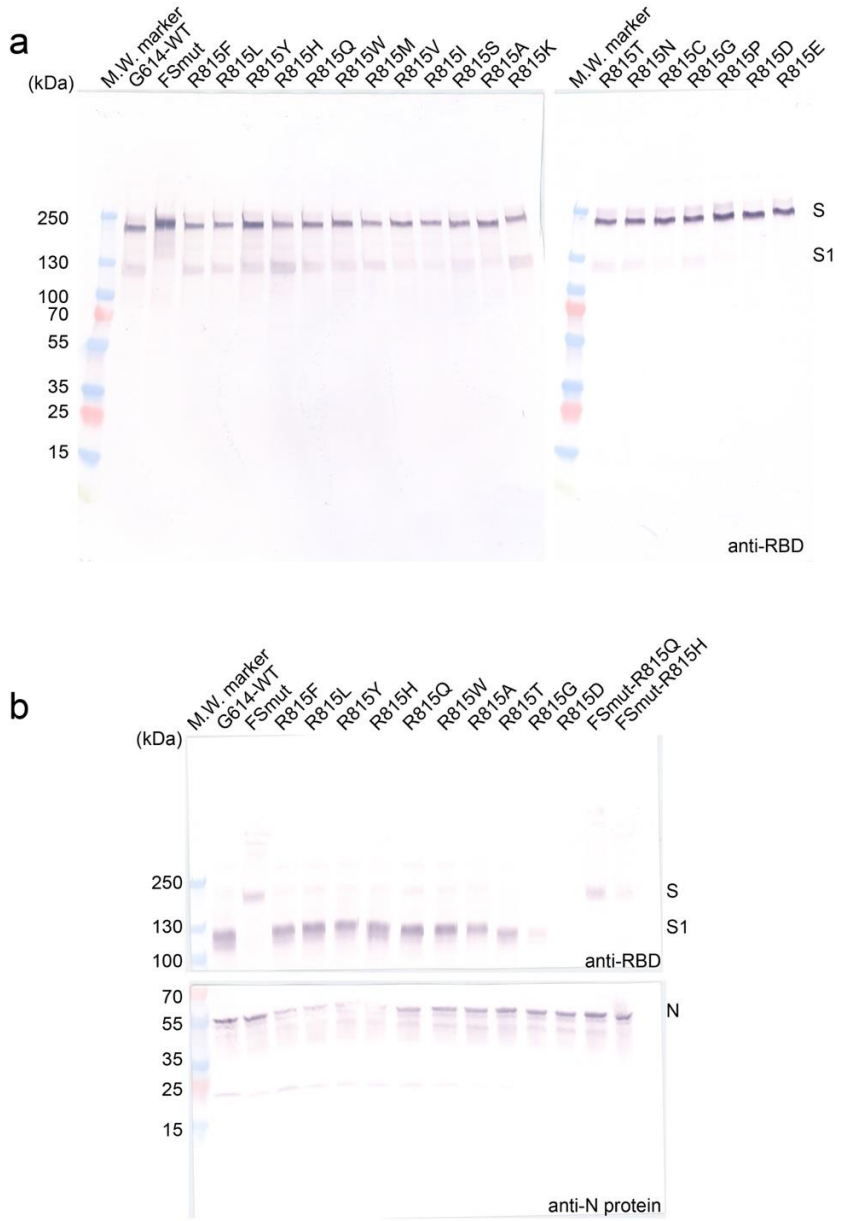
Supplementary Figure 5. TM helices in the S2' trimer. A close-up view of the three TM helices in the S2' trimer in detergent micelle. All the residues are shown in stick model.



Supplementary Figure 6. Effect of mutations in $S^{816-834}$ on membrane fusion. (a) Structure-guided design of mutations in $S^{816–834}$. The wildtype sequences are shown in black and the mutated residues in red. Mut-1 was described in our previous work[1]. (**b**) Cell–cell fusion mediated by the G614 S and the designed mutants. HEK293T cells transfected with the full-length S protein expression plasmids were fused with ACE2-expressing cells. Cell-cell fusion led to reconstitution of α and ω fragments of β-galactosidase to form an active enzyme, and the fusion activity was then quantified by a chemiluminescent assay. The experiment was repeated twice with similar results. Error bars, mean ± s.d..

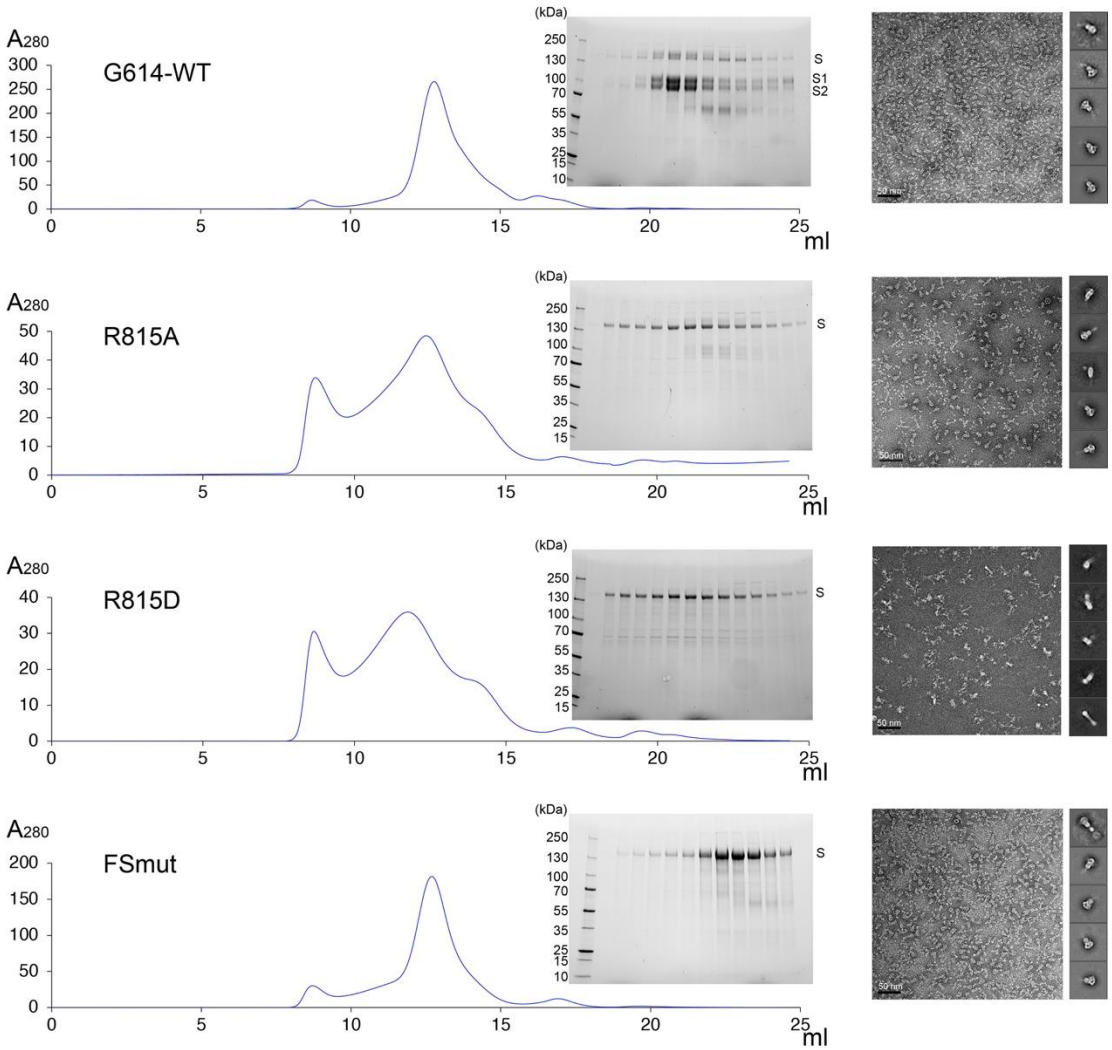


Supplementary Figure 7. Local structures near the S2' cleavage site in SARS-CoV-2 and SARS-CoV prefusion spike trimers. The prefusion trimer structures of SARS-CoV-2 (PDB ID:7KRQ[2]) and SARS-CoV (PDB ID:5X58[3]) are shown in ribbon diagram. The residues near the S2' cleavage site (Arg815 in SARS-CoV-2 and Arg797 in SARS-CoV) are indicated.

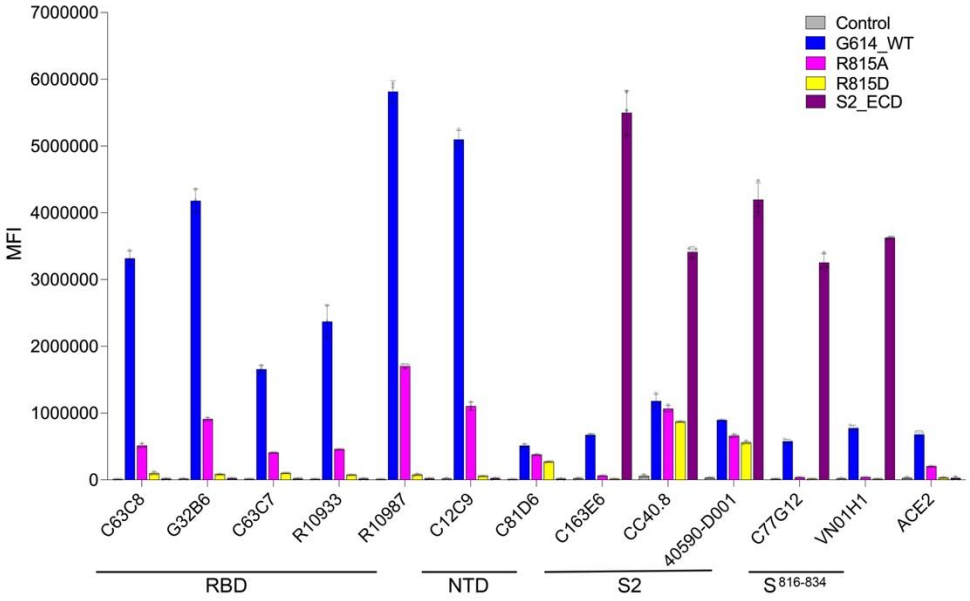


Supplementary Figure 8. Expression of the mutant spike proteins and their incorporation into SARS-CoV-2 VLPs. (**a**) Expression and processing of the full-length G614 S construct and its mutants in HEK293 cells. S protein samples prepared from HEK293 cells transiently transfected with 10 μg of the full-length S expression plasmids were detected by anti-RBD polyclonal antibodies. Bands for the uncleaved S and S1 fragment are indicated. The experiment was repeated twice with similar results. (**b**) VLP samples using G614 S construct and selected mutants were detected by anti-RBD polyclonal antibodies and an anti-N (nucleocapsid) protein antibody. Bands for the uncleaved S, S1 fragment and N protein are indicated. The experiment was repeated twice with similar results.

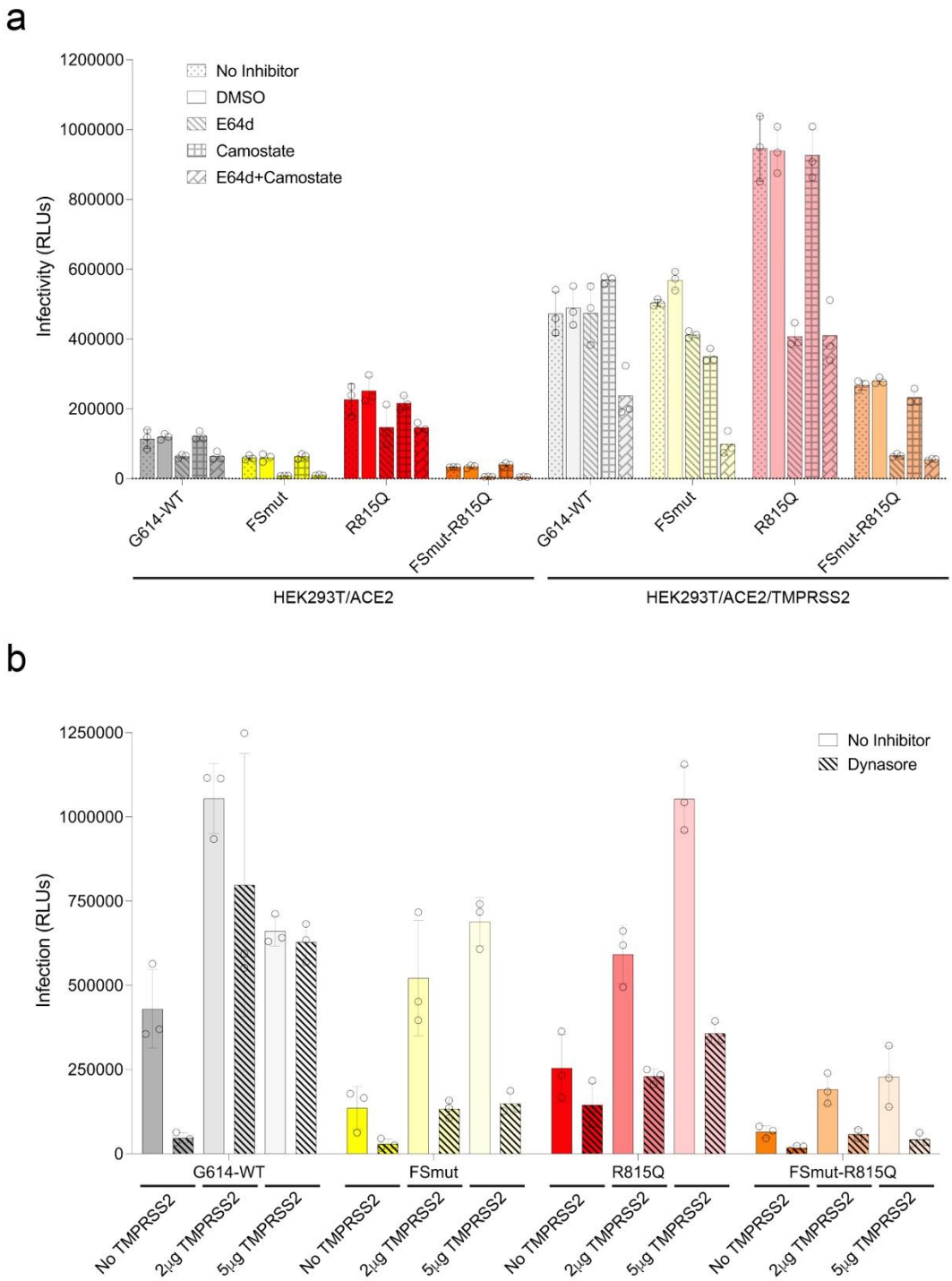
a


G614-WT
R815A
R815D
FSmut
A280
ml
(kDa)
S
S1
S2
50 nm

b

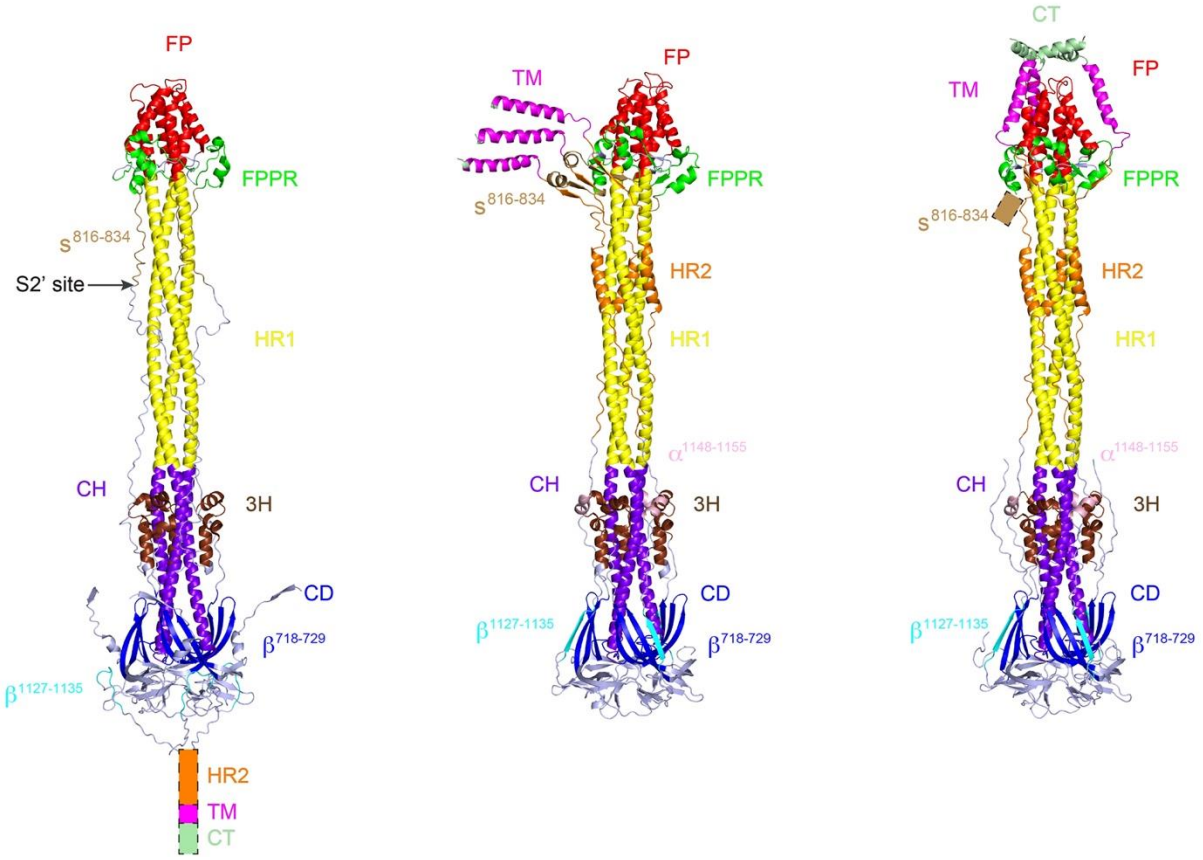

MFI
Control
G614_WT
R815A
R815D
S2_ECD
C63C8
G32B6
C63C7
R10933
R10987
C12C9
C81D6
C163E6
CC40.8
40590-D001
C77G12
VN01H1
ACE2
RBD
NTD
S2
S816-834

Supplementary Figure 9. Characterization of mutant spike proteins. (**a**) The purified full-length SARS-CoV-2 S proteins with mutations as indicated were resolved by gel-filtration chromatography on a Superose 6 column. Peak fractions were analyzed by SDS-PAGE. Representative 2D averages by negative stain EM of the peak fractions are also shown. The box size of 2D averages is ~460Å. G614-WT, R815A and FSmut showed predominantly the prefusion trimers, while R815D showed the postfusion trimers without any recognizable prefusion trimers. (**b**) Antibody binding to the full-length G614 S protein and its mutants R815A and R815D, as well as an S2 construct (S2_ECD) expressed on the cell surfaces analyzed by flow cytometry. The antibodies and their targets are indicated. A designed ACE2-based fusion inhibitor $ACE2_{615}$-foldon-T27W was used for detecting receptor binding[4]. MFI, mean fluorescent intensity. The error bars represent standard errors of mean from measurements using three independently transfected cell samples. The flow cytometry assays were repeated three times with essentially identical results. For the gating strategy, see Supplementary Figure 12

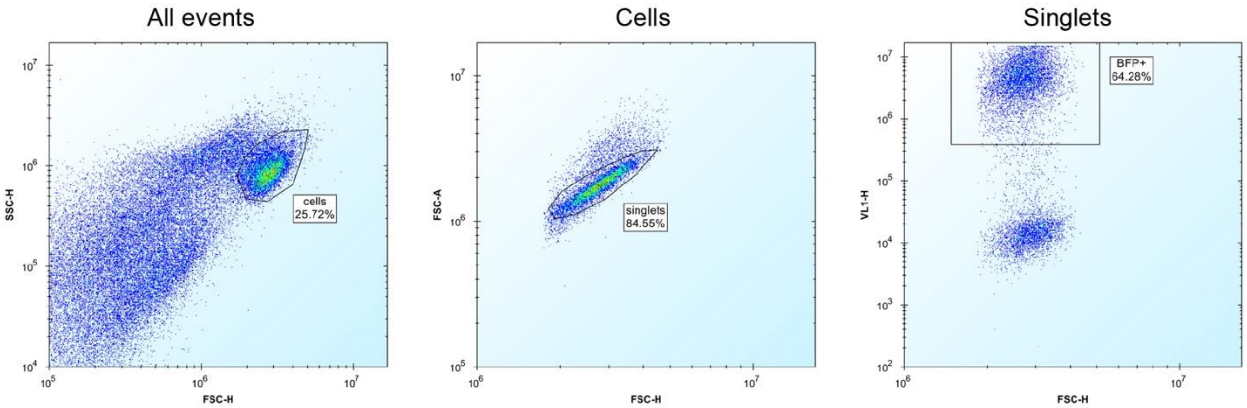


Supplementary Figure 10. Functional roles of host proteases that cleave the S2' site. (**a**) HEK293T cells transfected with the full-length S protein (G614) and selected mutants were tested for VLP infection with the ACE2-expressing cells or ACE2- and TMPRSS2-expressing cells, in the presence of cathepsin L inhibitor E64-d or TMPRSS2 inhibitor camostat, or both. DMSO was used for dissolving E64-d and also tested as a control. (**b**) HEK293T cells transfected with the full-

length S protein (G614) and selected mutants were tested for VLP infection with the ACE2-expressing cells or ACE2- and TMPRSS2-expressing cells, in the presence of the endocytosis inhibitor dynasore. The experiments were repeated at least twice with similar results. Error bars, mean ± s.d..



Supplementary Figure 11. Structural comparison of the early fusion intermediate S2 trimer, S2' trimer and uncleaved postfusion S2 trimer. Structures of the early fusion intermediate S2 trimer (PDB ID: 8Z7P[5]), S2' trimer (this study) and uncleaved postfusion S2 trimer (PDB ID: 8FDW[1]) are shown in ribbon diagram. Various structural components include $\beta^{718\text{-}729}$, 3H, $s^{816\text{-}834}$, FPPR, FP, HR1, CH, CD, $\beta^{1127\text{-}1135}$, $\alpha^{1148\text{-}1155}$, HR2, and CT.



Supplementary Figure 12. Gating strategies for flow cytometry experiments.

Supplementary Table 1. Cryo-EM data collection, refinement and validation statistics

	SARS-CoV-2 S2' protein (EMD-49912) (PDB ID 9NXY)	SARS-CoV-2 S2'-localmap1 (EMD-49917)	SARS-CoV-2 S2'-localmap2 (EMD-49918)
Data collection and processing			
Magnification	165,000	165,000	165,000
Voltage (kV)	300	300	300
Electron exposure (e–/Å^2)	51.929	51.929	51.929
Defocus range (μm)	-0.6 to -2.2	-0.6 to -2.2	-0.6 to -2.2
Pixel size (Å)	0.736	0.736	0.736
Symmetry imposed	C1	C1	C1
Initial particle images (no.)	902,903		
Final particle images (no.)	153,848	153,848	153,848
Map resolution (Å)	3.08	3.04	2.82
FSC threshold	0.143	0.143	0.143
Final map sampling (Å/pix)	0.736	0.736	0.736
Refinement			
Initial model used (PDB code)	8FDW		
Model resolution (Å)	3.3		
FSC threshold	0.5		
Model resolution range (Å)			
Map sharpening *B* factor (Å^2)			
Model composition			
Non-hydrogen atoms	11992		
Protein residues	1463		
Ligands	NAG:39, MAN:15		
B factors (Å^2)			
Protein (min/max/mean)	29.46/227.90/88.24		
Ligand (min/max/mean)	96.08/240.05/166.69		
R.M.S. deviations			
Bond lengths (Å)	0.005		
Bond angles (°)	0.495		
Validation			
MolProbity score	1.70		
Clashscore	5.97		
Poor rotamers (%)	1.92		
Ramachandran plot			
Favored (%)	97.11		
Allowed (%)	2.89		
Disallowed (%)	0.00		

References

1 Shi, W. *et al.* Cryo-EM structure of SARS-CoV-2 postfusion spike in membrane. *Nature* **619**, 403-409, doi:10.1038/s41586-023-06273-4 (2023).

2 Zhang, J. *et al.* Structural impact on SARS-CoV-2 spike protein by D614G substitution. *Science* **372**, 525-530, doi:10.1126/science.abf2303 (2021).

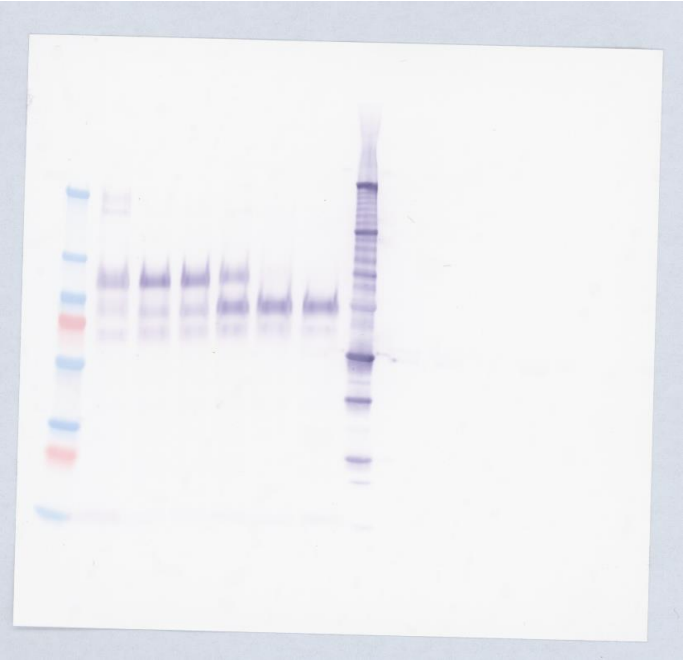
3 Yuan, Y. *et al.* Cryo-EM structures of MERS-CoV and SARS-CoV spike glycoproteins reveal the dynamic receptor binding domains. *Nat Commun* **8**, 15092, doi:10.1038/ncomms15092 (2017).

4 Xiao, T. *et al.* A trimeric human angiotensin-converting enzyme 2 as an anti-SARS-CoV-2 agent. *Nat Struct Mol Biol* **28**, 202-209, doi:10.1038/s41594-020-00549-3 (2021).

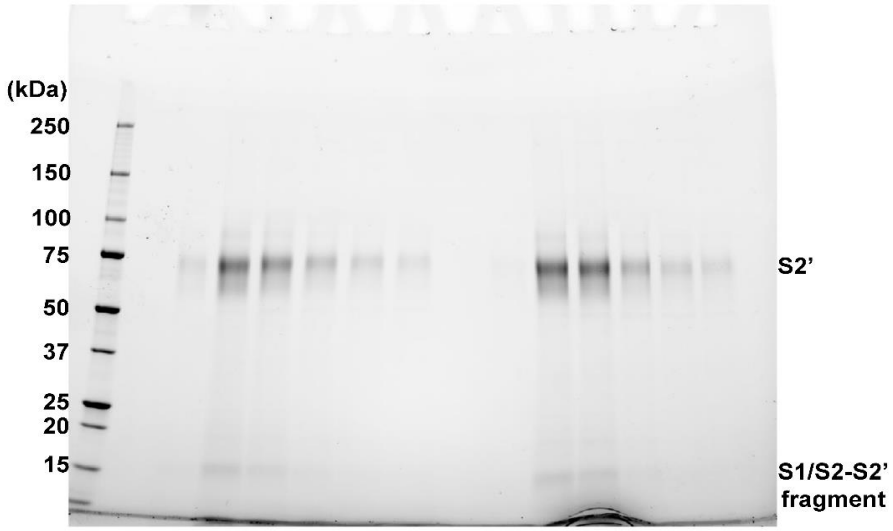
5 Xing, L. *et al.* Early fusion intermediate of ACE2-using coronavirus spike acting as an antiviral target. *Cell* **188**, 1297-1314 e1224, doi:10.1016/j.cell.2025.01.012 (2025).

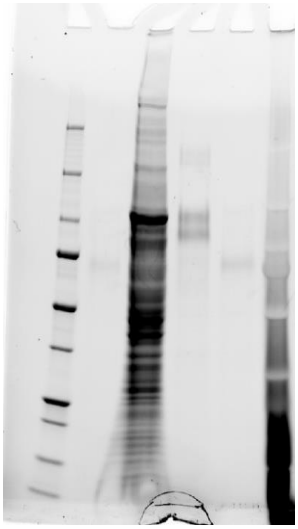
Uncropped scans of blots and gels in Supplementary Figures

1. Western blot for Supplementary Fig. 1b

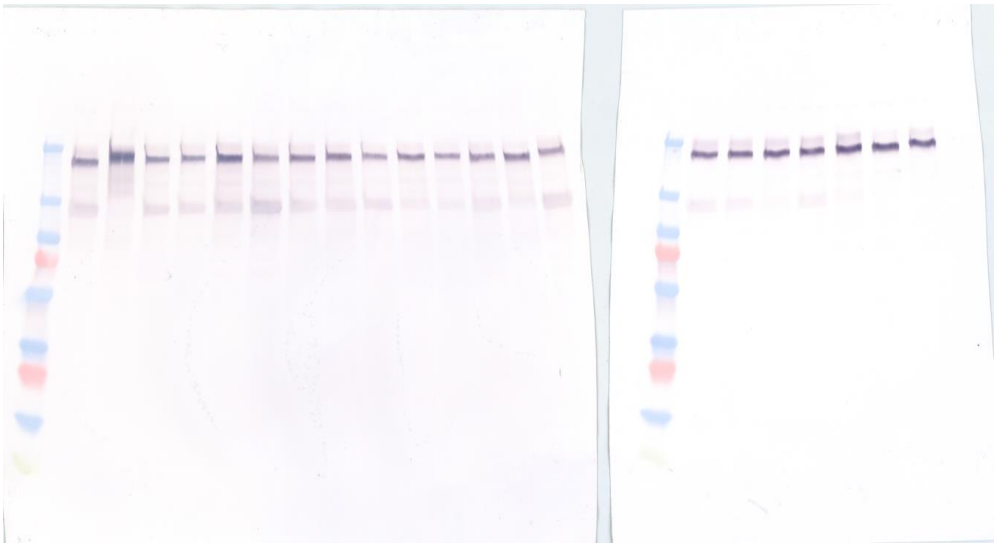


2. SDS-PAGE for Supplementary Fig. 1c





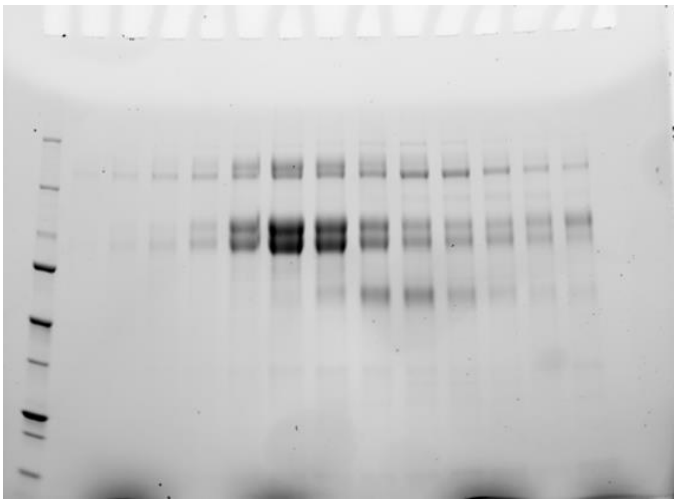
3. Western blot for Supplementary Fig. 8a



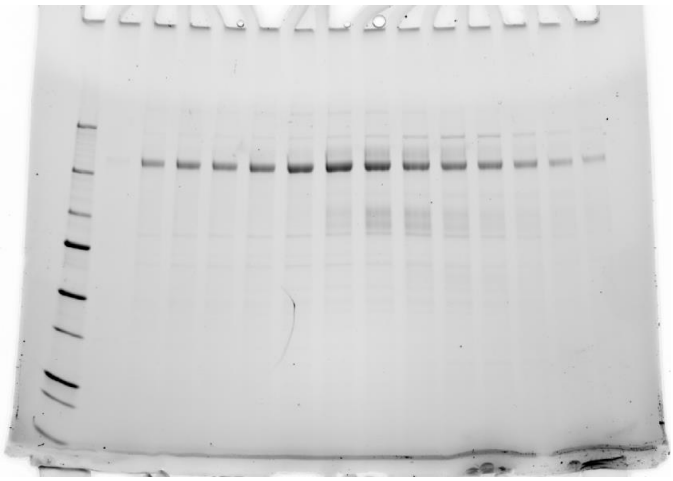
4. Western blot for Supplementary Fig. 8b

5. SDS-PAGE for Supplementary Fig. 9a

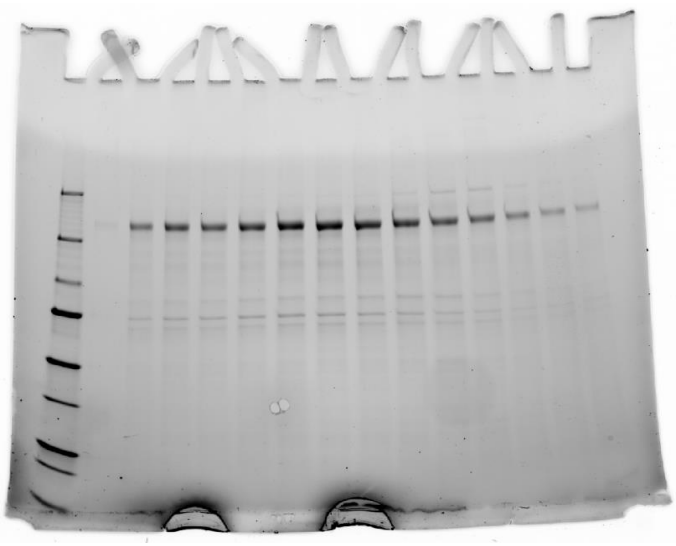
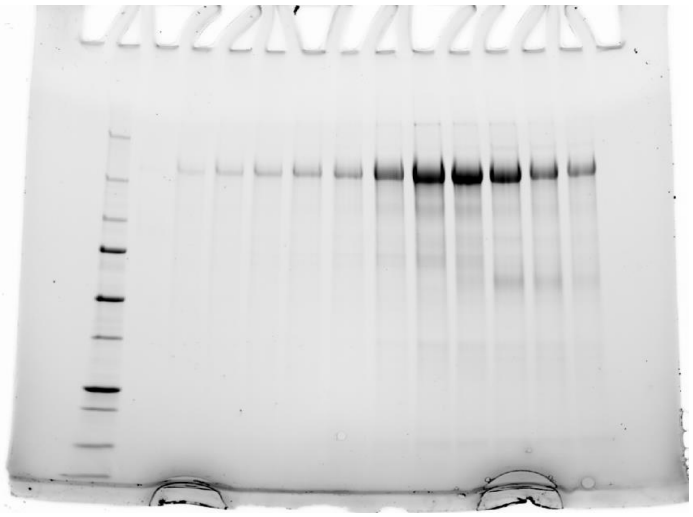
G614-WT



R815A



R815D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